UNIVERSIDAD AUTONOMA DE NUEVO LEON

Facultad de Ciencias Físico Matemáticas

(FCFM)

Unidad de Aprendizaje:

Desarrollo web – Tecnologías Servidor

Unidad 06

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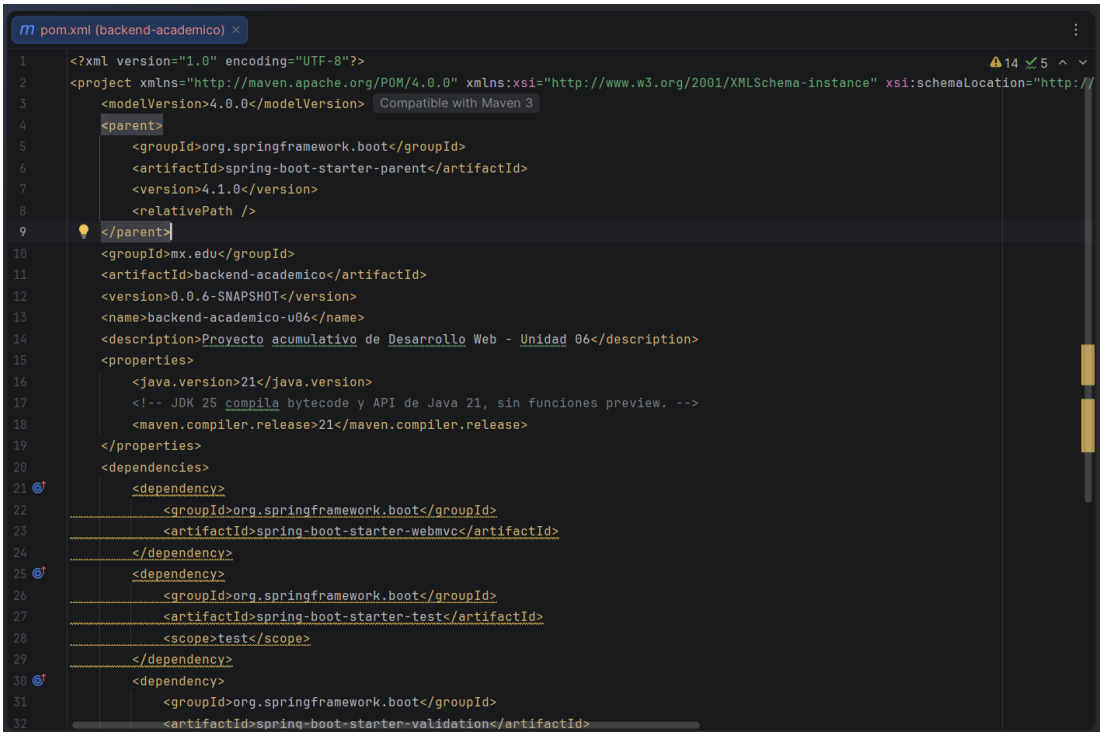
Grupo: 031

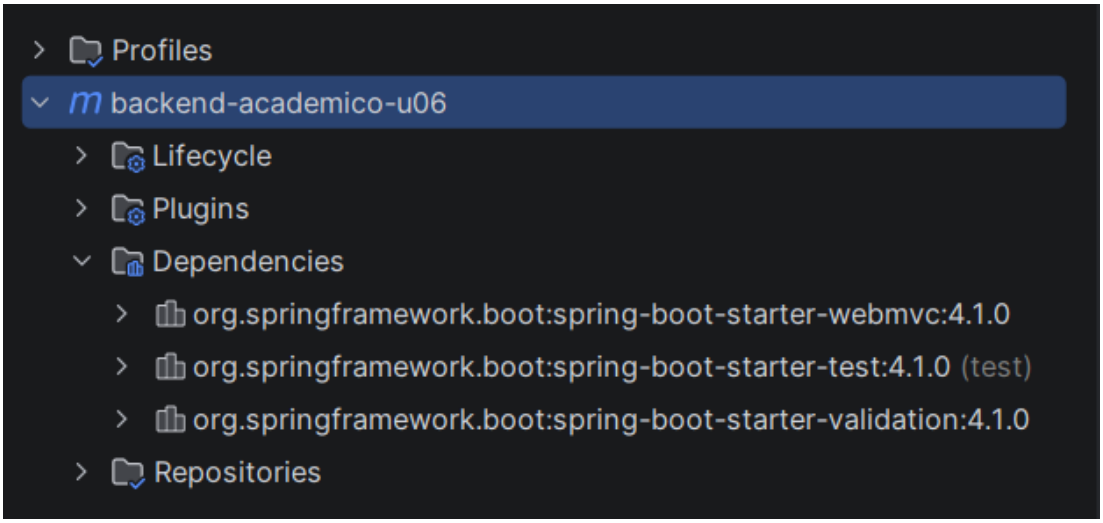
Profesor: Gustavo Adolfo Rodríguez Martínez

Carrera: LCC

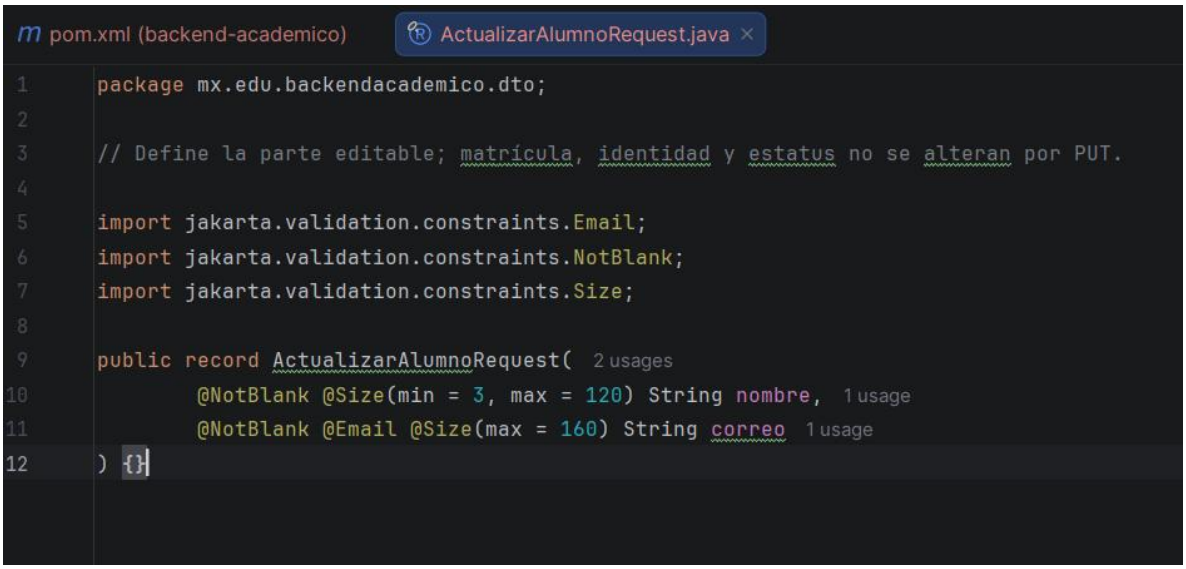
Monterrey N.L. a; 30 de octubre de 2026

Acción: modificar pom.xml.





Acción: modificar src/main/java/mx/edu/backendacademico/dto/ActualizarAlumnoRequest.java.



Acción: modificar src/main/java/mx/edu/backendacademico/dto/CrearAlumnoRequest.java.



Acción: modificar src/main/java/mx/edu/backendacademico/controller/AlumnoController.java.

AlumnoController.java

```java
package mx.edu.backendacademico.controller;

// Traduce HTTP a llamadas de aplicación y construye respuestas mediante DTO.

import jakarta.validation.Valid;
import java.net.URI;
import java.util.List;
import mx.edu.backendacademico.dto.ActualizarAlumnoRequest;
import mx.edu.backendacademico.dto.AlumnoMapper;
import mx.edu.backendacademico.dto.AlumnoResponse;
import mx.edu.backendacademico.dto.CrearAlumnoRequest;
import mx.edu.backendacademico.service.AlumnoService;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.DeleteMapping;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.PutMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;

@RestController
@RequestMapping("/api/v1/alumnos")
public class AlumnoController {
    private final AlumnoService service;  6 usages
    public AlumnoController(AlumnoService service) { this.service = service; }  no usages

    // GET consulta sin solicitar cambios de estado de negocio.
    @GetMapping
    public List<AlumnoResponse> listar() { return service.listar().stream().map(AlumnoMapper::toResponse).toList(); }

    // GET consulta sin solicitar cambios de estado de negocio.
    @GetMapping("/{id}")
    public AlumnoResponse buscar(@PathVariable Long id) { return AlumnoMapper.toResponse(service.buscar(id)); }

    // El servicio asigna identidad; ResponseEntity publica 201 y Location.
    @PostMapping
    public ResponseEntity<AlumnoResponse> crear(@RequestBody @Valid CrearAlumnoRequest request) {
        var creado = service.crear(request.matricula(), request.nombre(), request.correo());
        return ResponseEntity.created(URI.create("/api/v1/alumnos/" + creado.id())).body(AlumnoMapper.toResponse(creado));
    }
```

```java
    public AlumnoResponse buscar(@PathVariable Long id) { return AlumnoMapper.toResponse(service.buscar(id)); }

    // El servicio asigna identidad; ResponseEntity publica 201 y Location.
    @PostMapping
    public ResponseEntity<AlumnoResponse> crear(@RequestBody @Valid CrearAlumnoRequest request) {
        var creado = service.crear(request.matricula(), request.nombre(), request.correo());
        return ResponseEntity.created(URI.create("/api/v1/alumnos/" + creado.id())).body(AlumnoMapper.toResponse(creado));
    }

    // PUT sustituye todos los campos de la representación editable.
    @PutMapping("/{id}")
    public AlumnoResponse actualizar(@PathVariable Long id, @RequestBody @Valid ActualizarAlumnoRequest request) {
        return AlumnoMapper.toResponse(service.actualizar(id, request.nombre(), request.correo()));
    }

    // DELETE exitoso termina con 204, que no permite cuerpo de respuesta.
    @DeleteMapping("/{id}")
    public ResponseEntity<Void> eliminar(@PathVariable Long id) {
        service.eliminar(id);
        return ResponseEntity.noContent().build();
    }
}
```

Acción: crear src/main/java/mx/edu/backendacademico/exception/GlobalExceptionHandler.java.

GlobalExceptionHandler.java

```java
package mx.edu.backendacademico.exception;

import java.net.URI;
import java.util.List;
import java.util.UUID;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.http.*;
import org.springframework.web.bind.MethodArgumentNotValidException;
import org.springframework.web.bind.annotation.*;
import org.springframework.web.context.request.WebRequest;
import org.springframework.web.servlet.mvc.method.annotation.ResponseEntityExceptionHandler;

@RestControllerAdvice
public class GlobalExceptionHandler extends ResponseEntityExceptionHandler {
    private static final Logger log = LoggerFactory.getLogger(GlobalExceptionHandler.class); 1 usage

    @ExceptionHandler(AlumnoNoEncontradoException.class) no usages
    ProblemDetail noEncontrado(AlumnoNoEncontradoException ex) {
        return problema(HttpStatus.NOT_FOUND, codigo: "ALUMNO_NO_ENCONTRADO", ex.getMessage());
    }

    @ExceptionHandler(MatriculaDuplicadaException.class) no usages
    ProblemDetail conflicto(MatriculaDuplicadaException ex) {
        return problema(HttpStatus.CONFLICT, codigo: "MATRICULA_DUPLICADA", ex.getMessage());
    }

    @Override 1 usage
    protected ResponseEntity<Object> handleMethodArgumentNotValid(
            MethodArgumentNotValidException ex, HttpHeaders headers,
            HttpStatusCode status, WebRequest request) {
        var detalle = problema(HttpStatus.BAD_REQUEST, codigo: "DATOS_INVALIDOS", mensaje: "Revisa los campos indicados");
```

GlobalExceptionHandler.java

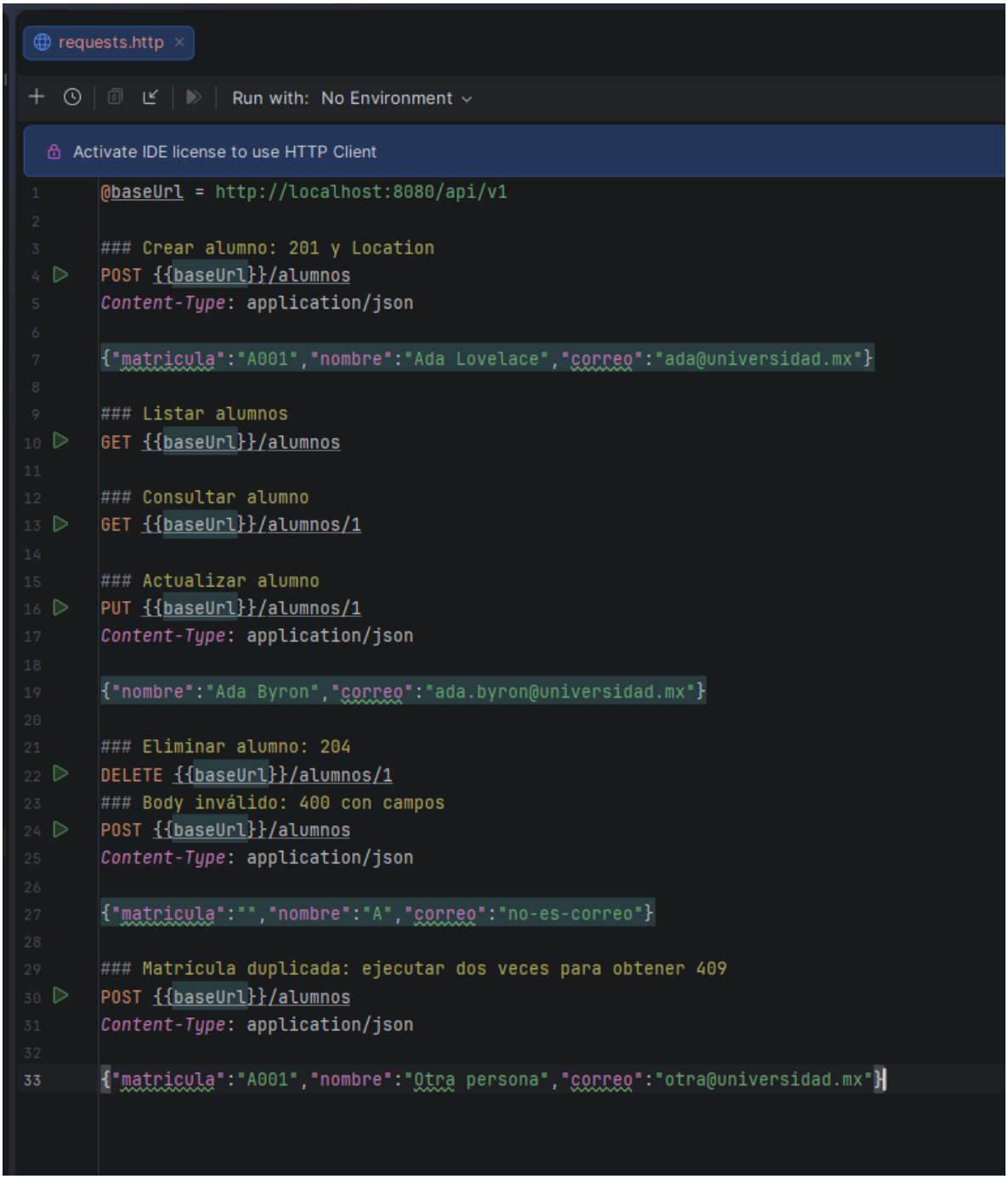
```java
public class GlobalExceptionHandler extends ResponseEntityExceptionHandler {
    protected ResponseEntity<Object> handleMethodArgumentNotValid(
        var detalle = problema(HttpStatus.BAD_REQUEST, codigo: "DATOS_INVALIDOS", mensaje: "Revisa los campos indicados");
        // Conserva varios errores de un campo sin publicar el valor rechazado.
        List<CampoInvalido> campos = ex.getBindingResult().getFieldErrors().stream()
                .map( FieldError e -> new CampoInvalido(e.getField(), e.getDefaultMessage())).toList();
        detalle.setProperty("campos", campos);
        return handleExceptionInternal(ex, detalle, headers, status, request);
    }

    @ExceptionHandler(Exception.class) no usages
    ProblemDetail inesperado(Exception ex) {
        String referencia = UUID.randomUUID().toString();
        // Registra una referencia y el tipo, sin cuerpo HTTP, contraseñas ni tokens.
        log.error("Falla no prevista referencia={} tipo={}", referencia, ex.getClass().getName());
        var detalle = problema(HttpStatus.INTERNAL_SERVER_ERROR, codigo: "ERROR_INTERNO", mensaje: "No fue posible completar la
        detalle.setProperty("referencia", referencia);
        return detalle;
    }

    private ProblemDetail problema(HttpStatus status, String codigo, String mensaje) { 4 usages
        var detalle = ProblemDetail.forStatusAndDetail(status, mensaje);
        detalle.setTitle(status.getReasonPhrase());
        detalle.setType(URI.create("urn:backend-academico:problema:" + codigo));
        detalle.setProperty("codigo", codigo);
        return detalle;
    }

    public record CampoInvalido(String campo, String mensaje) { } 2 usages
}
```

Acción: modificar requests.http.



Pruebas

```
PS C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-servidor\Unidades\Unidad 06\solucion> Invoke-RestMethod -Uri "http://localhost:8080/api/v1/alumnos" -Method POST -ContentType "application/json" -Body '{"matricula":"A001","
nombre":"Ada Lovelace","correo":"ada@universidad.mx"}'


id        : 1
matricula : A001
nombre    : Ada Lovelace
correo    : ada@universidad.mx
estatus   : ACTIVO



PS C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-servidor\Unidades\Unidad 06\solucion> Invoke-RestMethod -Uri "http://localhost:8080/api/v1/alumnos" -Method GET


id        : 1
matricula : A001
nombre    : Ada Lovelace
correo    : ada@universidad.mx
estatus   : ACTIVO
```

```
PS C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-servidor\Unidades\Unidad 06\solucion> Invoke-RestMethod -Uri "http://localhost:8080/api/v1/alumnos/1" -Method GET


id        : 1
matricula : A001
nombre    : Ada Lovelace
correo    : ada@universidad.mx
estatus   : ACTIVO
```

```
PS C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-servidor\Unidades\Unidad 06\solucion> Invoke-RestMethod -Uri "http://localhost:8080/api/v1/alumnos/1" -Method PUT -ContentType "application/json" -Body '{"nombre":"Ada Byron
","correo":"ada.byron@universidad.mx"}'


id        : 1
matricula : A001
nombre    : Ada Byron
correo    : ada.byron@universidad.mx
estatus   : ACTIVO
```

```
PS C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-servidor\Unidades\Unidad 06\solucion> Invoke-RestMethod -Uri "http://localhost:8080/api/v1/alumnos" -Method POST -ContentType "application/json" -Body '{"matricula":"","nom
re":"A","correo":"no-es-correo"}'
Invoke-RestMethod : Error en el servidor remoto: (400) Solicitud incorrecta.
En línea: 1 Carácter: 1
+ Invoke-RestMethod -Uri "http://localhost:8080/api/v1/alumnos" -Method ...
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : InvalidOperation: (System.Net.HttpWebRequest:HttpWebRequest) [Invoke-RestMethod], WebException
    + FullyQualifiedErrorId : WebCmdletWebResponseException,Microsoft.PowerShell.Commands.InvokeRestMethodCommand
```

```
PS C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-servidor\Unidades\Unidad 06\solucion> Invoke-RestMethod -Uri "http://localhost:8080/api/v1/alumnos" -Method POST -ContentType "application/json" -Body '{"matricula":"A001","
nombre":"Otra persona","correo":"otra@universidad.mx"}'
Invoke-RestMethod : Error en el servidor remoto: (409) Conflicto.
En línea: 1 Carácter: 1
+ Invoke-RestMethod -Uri "http://localhost:8080/api/v1/alumnos" -Method ...
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : InvalidOperation: (System.Net.HttpWebRequest:HttpWebRequest) [Invoke-RestMethod], WebException
    + FullyQualifiedErrorId : WebCmdletWebResponseException,Microsoft.PowerShell.Commands.InvokeRestMethodCommand
```

```
PS C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-servidor\Unidades\Unidad 06\solucion> Invoke-RestMethod -Uri "http://localhost:8080/api/v1/alumnos/1" -Method DELETE


PS C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-servidor\Unidades\Unidad 06\solucion> Invoke-RestMethod -Uri "http://localhost:8080/api/v1/alumnos/1" -Method GET
Invoke-RestMethod : Error en el servidor remoto: (404) No se encontró.
En línea: 1 Carácter: 1
+ Invoke-RestMethod -Uri "http://localhost:8080/api/v1/alumnos/1" -Meth ...
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : InvalidOperation: (System.Net.HttpWebRequest:HttpWebRequest) [Invoke-RestMethod], WebException
    + FullyQualifiedErrorId : WebCmdletWebResponseException,Microsoft.PowerShell.Commands.InvokeRestMethodCommand
```

Comprobación del resultado

Envía una matrícula vacía, un nombre de un carácter y un correo inválido: espera 400 con codigo=DATOS_INVALIDOS y una lista de campos. Envía JSON incompleto: espera 400 por lectura del mensaje. Consulta un id inexistente: espera 404. Crea dos veces la misma matrícula: espera 409 en la segunda petición. Vuelve a ejecutar las pruebas antes de pasar a persistencia.





Json incompleto:



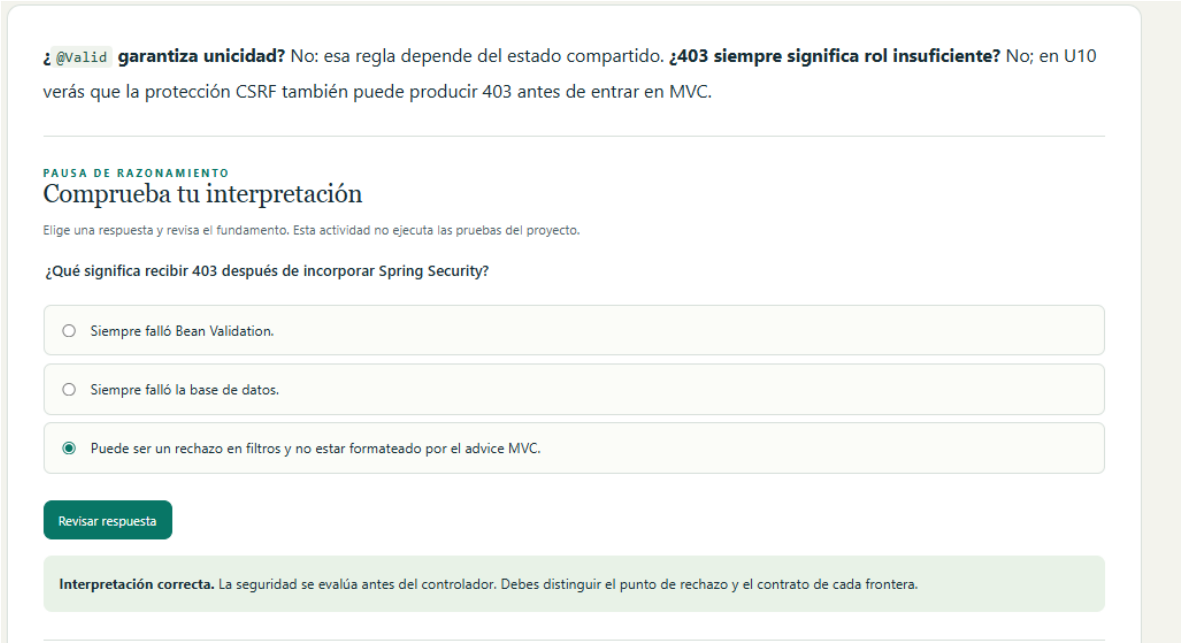
Consulta de id:



Matricula duplicada:

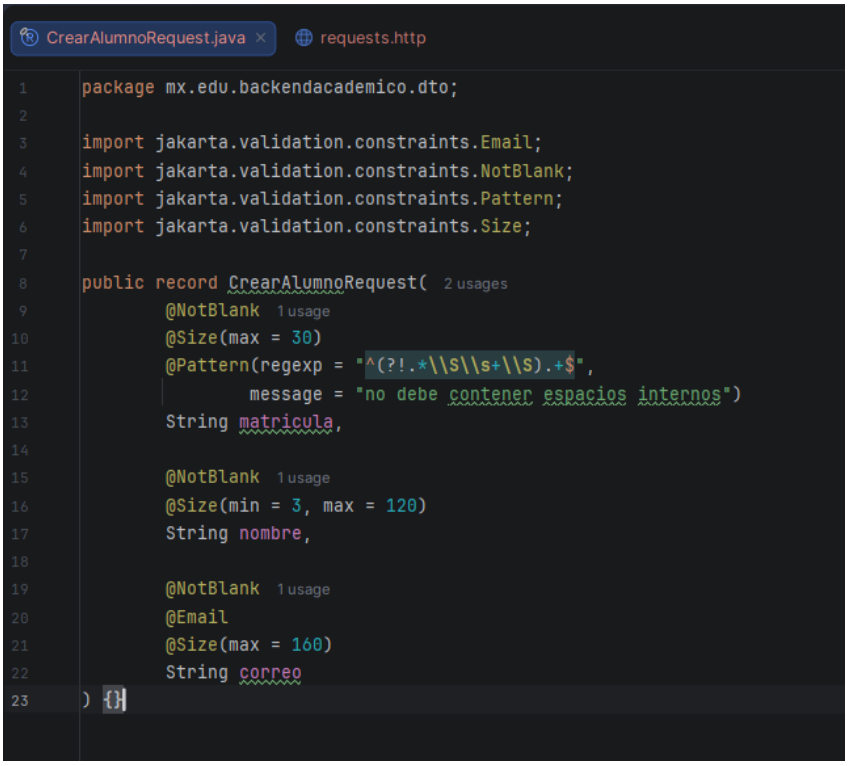


Autoevaluación razonada

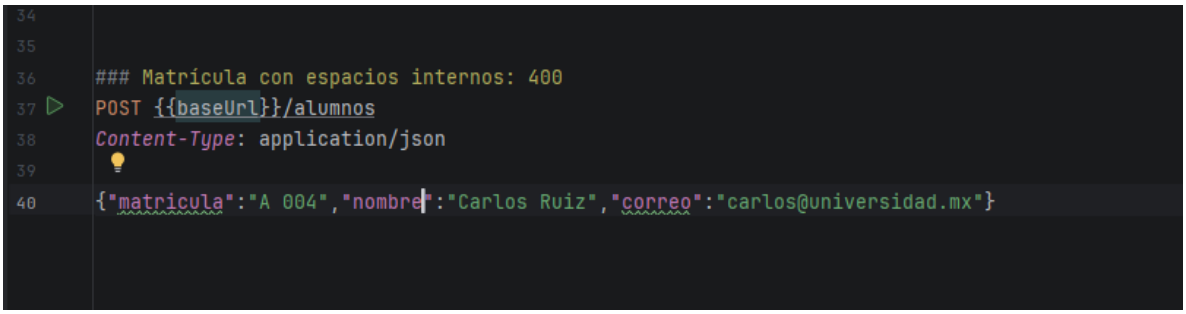


Reto

Agrega una regla declarativa que rechace una matrícula con espacios internos y una prueba HTTP para ella. Conserva los límites y la normalización existentes. Explica qué entradas rechaza la frontera y qué representación canónica guarda el dominio. No cambies los códigos de error ya publicados.



Request



```
PS C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-
servidor\Unidades\Unidad 06\reto> $body = @{
>>     matricula = "A 004"
>>     nombre = "Carlos Ruiz"
>>     correo = "carlos@universidad.mx"
>> } | ConvertTo-Json -Compress
PS C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-
servidor\Unidades\Unidad 06\reto>
PS C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-
servidor\Unidades\Unidad 06\reto> try {
>>     $r = Invoke-WebRequest -Uri "http://localhost:8080/api/v1/alumnos"
-Method POST -ContentType "application/json" -Body $body
>>     $r.StatusCode
>>     $r.Content
>> }
>> catch {
>>     [int]$_.Exception.Response.StatusCode
>>     $reader = New-Object
System.IO.StreamReader($_.Exception.Response.GetResponseStream())
>>     $reader.ReadToEnd()
>> }
400
{"detail":"Revisa los campos
indicados","instance":"/api/v1/alumnos","status":400,"title":"Bad
Request","type":"urn:backend-
academico:problema:DATOS_INVALIDOS","codigo":"DATOS_INVALIDOS","campos":[
{"campo":"matricula","mensaje":"no debe contener espacios internos"}]}
PS C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-
servidor\Unidades\Unidad 06\reto>
```

Se agregó una regla declarativa mediante @Pattern para impedir que las matrículas contengan espacios internos, conservando las restricciones de longitud y la normalización existente.

Se realizó una prueba HTTP enviando la matrícula A 004, obteniendo una respuesta 400 Bad Request con el código DATOS_INVALIDOS.

Finalmente, se ejecutaron nuevamente las pruebas de Maven, obteniendo BUILD SUCCESS, con 4 pruebas aprobadas y ningún error.

```
PS C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-servidor\Unidades\Unidad 06\reto> .\mvnw.cmd clean test
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner
(file:/C:/Users/andre/.m2/wrapper/dists/apache-maven-3.9.11/03d7e36a140982eea48e22c1dcac01d8862b2550b2939e09a0809bbc5182a5bc/lib/guice-5.1.0-
classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< mx.edu:backend-academico >----------------------
[INFO] Building backend-academico-u06 0.0.6-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ backend-academico ---
[INFO] Deleting C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-servidor\Unidades\Unidad 06\reto\target
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ backend-academico ---
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ backend-academico ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 18 source files with javac [debug parameters release 21] to target\classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ backend-academico ---
[INFO] skip non existing resourceDirectory C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-servidor\Unidades\Unidad 06\reto\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ backend-academico ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 2 source files with javac [debug parameters release 21] to target\test-classes
[INFO]
[INFO] --- surefire:3.5.6:test (default-test) @ backend-academico ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running mx.edu.backendacademico.BackendAcademicoApplicationTests
23:54:33.652 [main] INFO org.springframework.test.context.support.AnnotationConfigContextLoaderUtils -- Could not detect default configuration
classes for test class [mx.edu.backendacademico.BackendAcademicoApplicationTests]: BackendAcademicoApplicationTests does not declare any static,
non-private, non-final, nested classes annotated with @Configuration.
23:54:33.796 [main] INFO org.springframework.boot.test.context.SpringBootTestContextBootstrapper -- Found @SpringBootConfiguration
mx.edu.backendacademico.BackendAcademicoApplication for test class mx.edu.backendacademico.BackendAcademicoApplicationTests
23:54:33.915 [main] INFO org.springframework.test.context.support.AnnotationConfigContextLoaderUtils -- Could not detect default configuration
classes for test class [mx.edu.backendacademico.BackendAcademicoApplicationTests]: BackendAcademicoApplicationTests does not declare any static,
non-private, non-final, nested classes annotated with @Configuration.
23:54:33.917 [main] INFO org.springframework.boot.test.context.SpringBootTestContextBootstrapper -- Found @SpringBootConfiguration
mx.edu.backendacademico.BackendAcademicoApplication for test class mx.edu.backendacademico.BackendAcademicoApplicationTests

  .
 /\\ / ___'_ __ _ _(_)_ __  __ _ \ \ \ \
( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
  '  |____| .__|_| |_|_| |_\__, | / / / /
 =========|_|==============|___/=/_/_/_/

 :: Spring Boot ::                (v4.1.0)

2026-10-30T23:54:34.371-06:00  INFO 31500 --- [backend-academico] [           main] m.e.b.BackendAcademicoApplicationTests   : Starting
BackendAcademicoApplicationTests using Java 24.0.2 with PID 31500 (started by andre in C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-
servidor\Unidades\Unidad 06\reto)
2026-10-30T23:54:34.373-06:00  INFO 31500 --- [backend-academico] [           main] m.e.b.BackendAcademicoApplicationTests   : No active profile
set, falling back to 1 default profile: "default"
2026-10-30T23:54:35.293-06:00  INFO 31500 --- [backend-academico] [           main] m.e.backendacademico.config.CicloDeVida  : Bean inicializado en
2026-10-31T05:54:35.293570900Z
2026-10-30T23:54:35.818-06:00  INFO 31500 --- [backend-academico] [           main] m.e.b.BackendAcademicoApplicationTests   : Started
BackendAcademicoApplicationTests in 1.814 seconds (process running for 3.203)
Mockito is currently self-attaching to enable the inline-mock-maker. This will no longer work in future releases of the JDK. Please add Mockito as
an agent to your build as described in Mockito's documentation: https://javadoc.io/doc/org.mockito/mockito-
core/latest/org.mockito/org/mockito/Mockito.html#0.3
WARNING: A Java agent has been loaded dynamically (C:\Users\andre\.m2\repository\net\bytebuddy\byte-buddy-agent\1.18.10\byte-buddy-agent-
1.18.10.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 3.213 s -- in mx.edu.backendacademico.BackendAcademicoApplicationTests
[INFO] Running mx.edu.backendacademico.domain.AlumnoTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.030 s -- in mx.edu.backendacademico.domain.AlumnoTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  8.220 s
[INFO] Finished at: 2026-10-30T23:54:36-06:00
[INFO] ------------------------------------------------------------------------
```